

MLIR NPU Compiler - Lecture 19 Worksheet

Runtime / ABI / Codegen

제출 산출물: tensor descriptor table, command buffer schema, runtime API sketch, codegen decision, FileCheck test plan.

Exercise A - Tensor Descriptor ABI

- Input A: shape [1, 4096], dtype int8, row-major, DRAM, per-tensor scale=0.03125, zero_point=0
- Weight B: shape [4096, 4096], dtype int4 packed, layout K-major/N-blocked, CONST memory
- Output C: shape [1, 4096], dtype int8, DRAM, per-tensor output scale=0.0625
- 각 tensor 에 대해 descriptor field 를 채우고, logical shape 와 physical stride 를 구분하세요.

Field	Your value	Reason	Check
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Exercise B - Command Buffer Packet

- DMA_LOAD A tile, DMA_LOAD B tile, WAIT, MATMUL, REQUANT, DMA_STORE, FENCE packet 을 작성하세요.
- 각 command 의 descriptor index, byte offset, byte count, token produce/wait field 를 표시하세요.
- 어떤 command 를 overlap 할 수 있는지 표시하세요.

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Exercise C - Lowering Path

- npu_rt.command_buffer 를 LLVM IR stub, EmitC C++ stub, binary artifact 중 하나로 lowering 하는 전략을 선택하세요.
- 선택 이유를 embedded deployment, debug, performance, firmware update 관점에서 설명하세요.

- FileCheck positive/negative test 를 각각 하나씩 설계하세요.

Field	Your value	Reason	Check
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Exercise D - Runtime Capability/Fallback

- Runtime capability query 결과가 native tile 16x16x32, supported dtype {int8, bf16}, max command buffer 1MB 라고 가정하세요.
- 모델이 int4 weight GEMM 을 요구할 때 compiler 가 선택할 fallback policy 를 정의하세요.
- compile-time failure, CPU fallback, int4->int8 unpack, firmware update requirement 중 어떤 선택이 타당한지 판단하세요.

Field	Your value	Reason	Check
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FileCheck Test Template

```
// RUN: mlir-opt %s --npu-rt-pack-command-buffer | FileCheck %s

// CHECK-LABEL: npu_rt.command_buffer @matmul_tile
// CHECK: command_count = 6
// CHECK: descriptor_count = 5
// CHECK: DMA_LOAD
// CHECK: MATMUL
// CHECK: REQUANT
// CHECK: DMA_STORE
// CHECK: FENCE

// NEGATIVE: expect diagnostic for unsupported layout or missing fence
```